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# The RTL-to-GDSII Flow with LibreLane



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**KV VLSI Design, 336.051, WiSe 26/27**

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# Today's Goals

After this lecture, you can

- run **LibreLane** and find every result in its run folder
- plan a **floorplan**: core, rows, power grid, pins
- explain **placement**, **buffering**, **clock tree**, **routing**
- read the **signoff**: corners, IR drop, DRC, LVS, LEC
- explain why estimates before routing are **optimistic**
- describe how a macro becomes a **chip** with a padframe

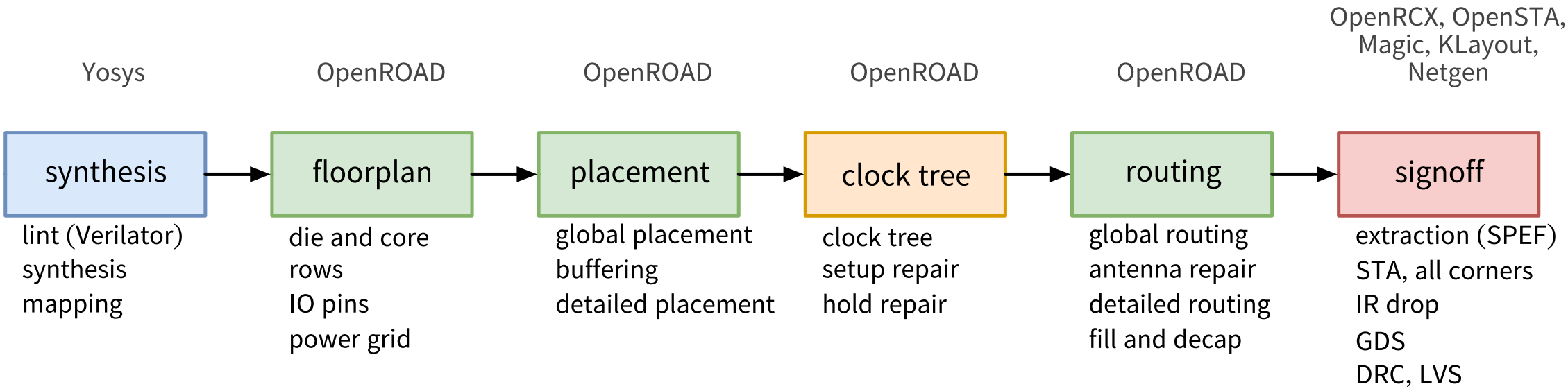
Chapter 10

Lecture notes: Chapter 10 of iic-jku.github.io/vlsi-design

# LibreLane



# From RTL to Layout




- Each **step** runs one tool on the state the last one left, **checkers** stop on errors
- **Classic** flow hardens a macro, **Chip** flow adds the padframe
- `runs/base/NN-step/`: logs, reports, and the exact `COMMANDS`
- `final/`: GDS, LEF, netlists, SPEF, SDF, metrics

# Configuring a Run

```
DESIGN_NAME: bs_reg
VERILOG_FILES: [.../barrel_shift.sv, .../bs_reg.sv]
CLOCK_PORT: clk_i
CLOCK_PERIOD: 4.0
PNR_SDC_FILE: dir::.../bs_reg/syn/bs_reg.sdc
SIGNOFF_SDC_FILE: dir::.../bs_reg/syn/bs_reg.sdc
FP_SIZING: relative
FP_CORE_UTIL: 50
PL_TARGET_DENSITY_PCT: 60
VDD_NETS: [VDD]
GND_NETS: [VSS]
STD_CELL_LIBRARY: sg13cmos5l_stdcell   # corners from the PDK
```

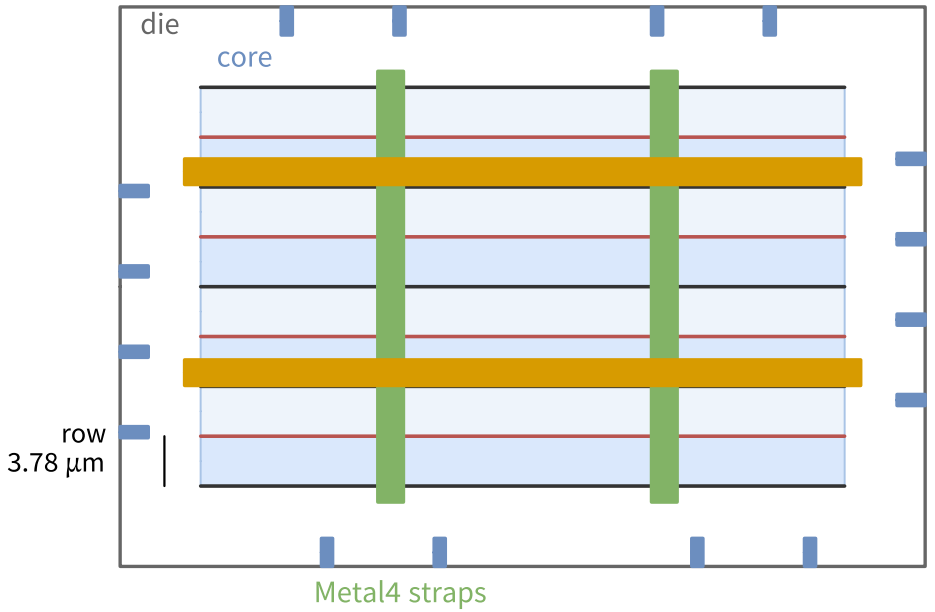
```
librelane config.yaml --pdk ihp-sg13cmos5l --pdk-root $PDK_ROOT --manual-pdk --run-tag base
```

- Everything not set comes from LibreLane and from `libs.tech/librelane/` of the PDK

# Floorplan



# Die, Core, Rows, and Power




Metal1 rails: VDD (red), VSS (black)

TopMetal1 straps

pins on Metal3 (left, right),
Metal2 (top, bottom)

- `FP_CORE_UTIL`: core = cell area / utilization
- Rows of **3.78 µm**, sites of **0.48 µm**
- No tap cells: the cells tie their wells
- Metal1 rails, straps on **Metal4** and **TopMetal1**, 50 µm pitch
- Pins on Metal2 and Metal3

**Exercise** (Section 10.3): plan the floorplan of bs_reg, and make the core denser.

# Placement



# Placement and Buffering

- **Global placement**: cells repel like charges, nets pull, target density
- STA **estimates** each wire: tree length × R and C per micron
- **Repair of the design**: slew, capacitance, fanout above **8**
  - a `buf_1` at each of the 40 inputs and 32 outputs
  - 142 more buffers split 30 high-fanout nets into trees
- **Detailed placement**: every cell onto a site of a row

# Clock Tree Synthesis



# Clock Tree Synthesis

- Until now an ideal clock, now a **tree**: equal **latency**, small **skew**
- TritonCTS: **H-tree** from the PDK clock buffers (`buf_8` ... `buf_1`)
- STA from now on: the **propagated** clock
- Timing repair: larger cells for setup, `dlygate` delay cells for hold
- Clock gate (`cg_mix`): CTS **balances** with 3 delay buffers, skew 0.06 ns
- Divided clock (`div_mix`): two trees, **no balancing**, 32 delay cells for hold

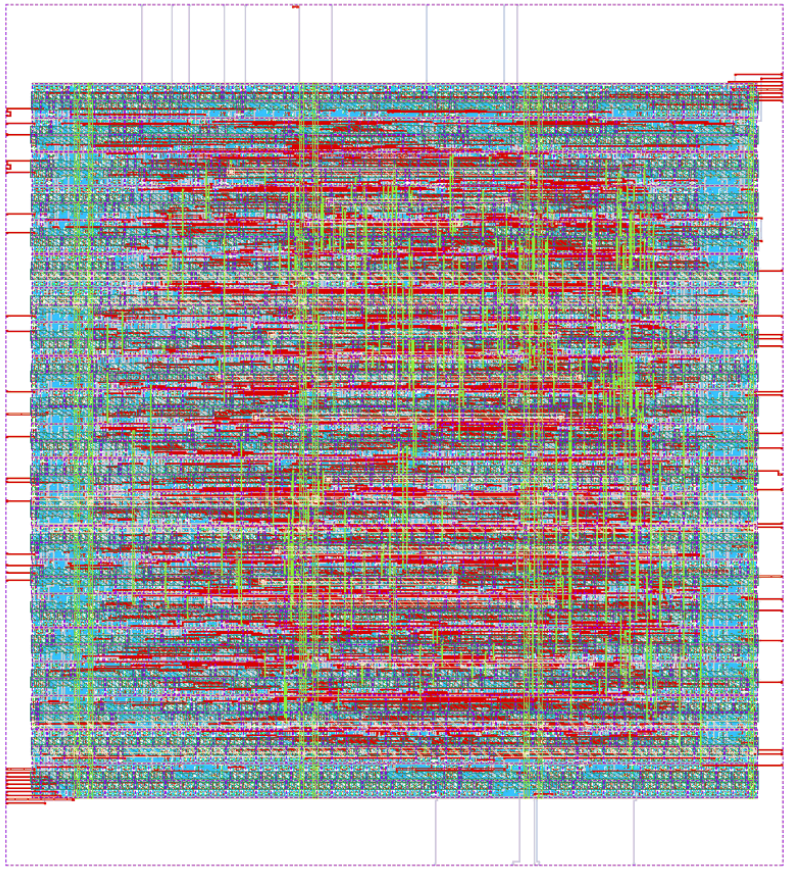
**Exercise** (Section 10.5): follow the clock and the reset net, and ask for more hold margin.

# Routing



# Routing and Fill

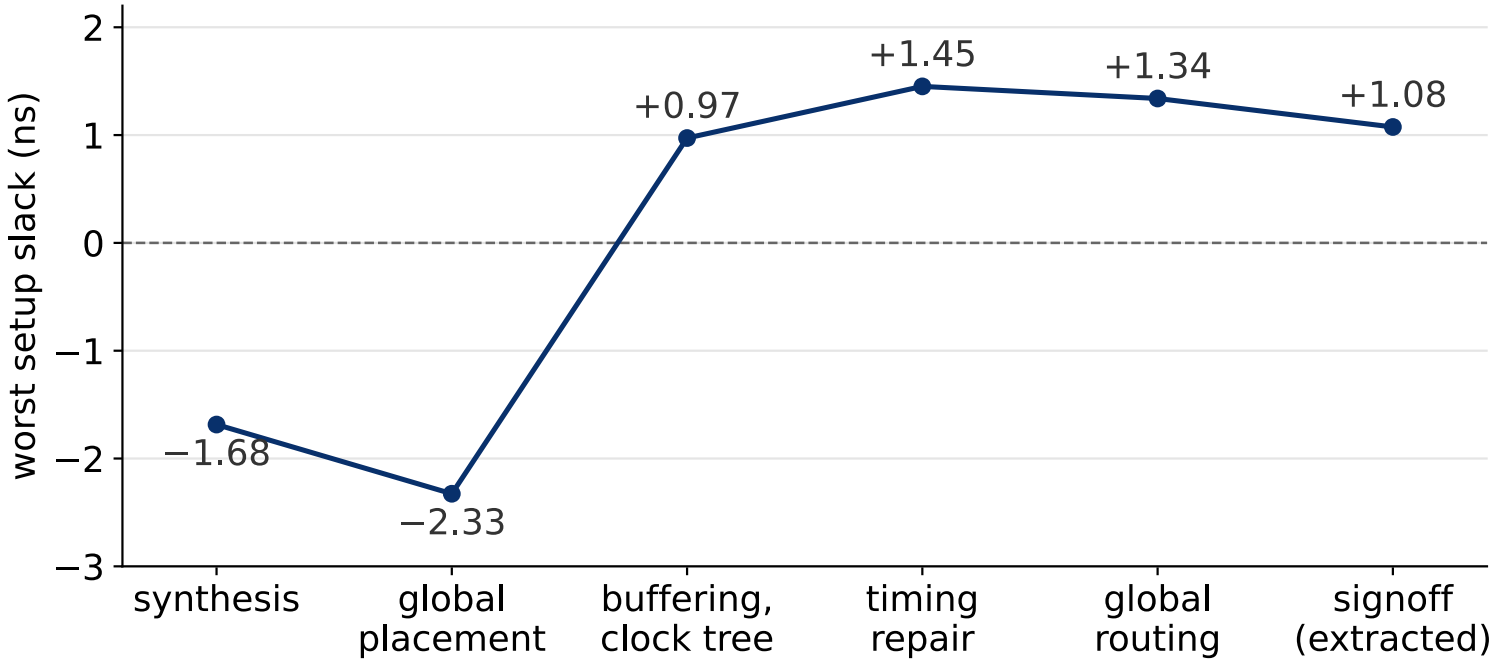
- **Global routing**: tiles, layers, congestion
  - 44 % of Metal2, 39 % of Metal3, no overflow
- **Antenna rule**: jumper or diode `antennanp`
- **Detailed routing**: repairs violations in iterations until none are left
- **Fill**: decaps and fillers in the gaps



# Signoff



# Extraction and Timing




- **OpenRCX** extracts R and C of every wire (SPEF), STA checks **all corners**
- Estimates before routing are **optimistic**: detours, vias, coupling

# Checking the Layout

- **IR drop** in the power grid: 1.33 mV worst, 0.11 % of 1.2 V
- **DRC** with Magic and KLayout, and an XOR of the GDS files from Magic and KLayout
- **LVS**: Magic extracts, Netgen compares, cells as abstracts
- **LEC** with EQY, run by hand: the final netlist against the RTL

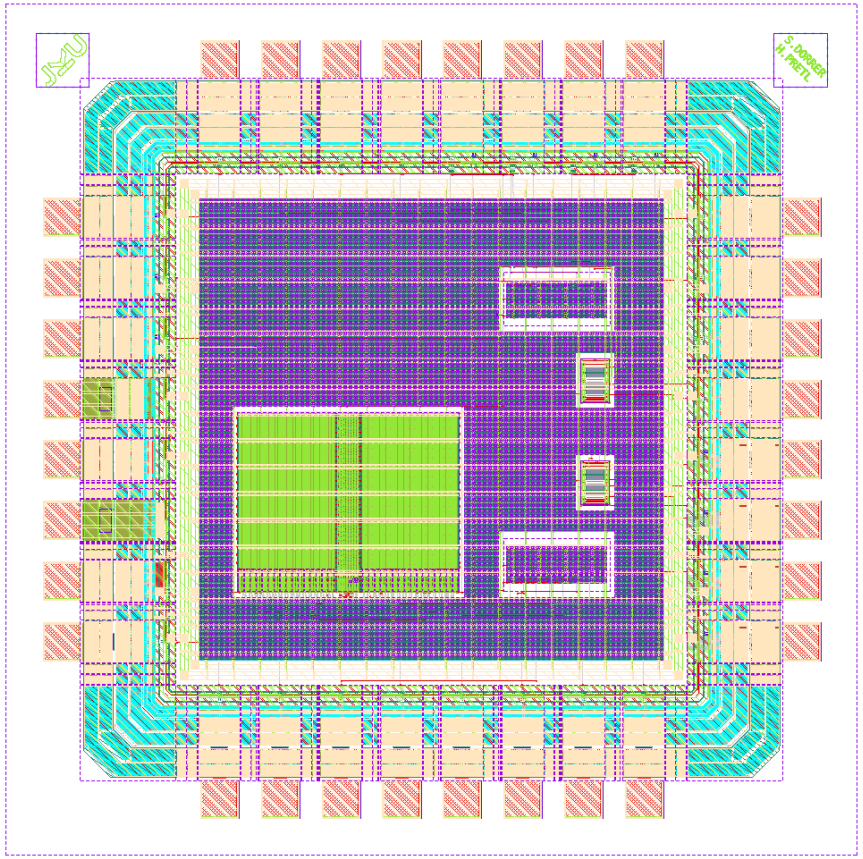
**Exercise** (Section 10.7): find what fails in signoff, close the timing, and prove the netlist.

# Chip Assembly



# From Macro to Chip

- AMS chip template: LibreLane's **Chip** flow
- **Padframe**: IO cells with bondpads, ESD, supplies
  - 1.6 × 1.6 mm, 32 pads in four lists, `PAD_WEST` to `PAD_EAST`
- **Seal ring** along the edge of the die
- `MACROS`: GDS, LEF, netlists, timing per macro
- Core ring to the supply pads, non-default rules for analog nets
- Then chip logo, fill, DRC, bonding diagram



# Summary

- LibreLane: **steps** with their own folders, results in `final/`
- Floorplan: core from **utilization**, rows, Metal1 rails and straps
- Placement buffers every net beyond **slew, cap, fanout** limits
- CTS: **H-tree**, propagated clock, setup and hold repair
- Routing: global, antenna, detailed, then **fill**
- Signoff: **extraction**, all corners, IR drop, DRC, LVS, then **LEC**
- Chip: macros in a core inside a **padframe** and a seal ring

# Exercises

1. Plan the Floorplan (10.3)
2. Follow the Clock and the Reset (10.5)
3. Close the Timing (10.7)

All code in `content/digital_flow/code` of the course repository, every number reproduced by `scripts/check_digital_flow.sh`



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